

Luminescence Postprocessing Pipeline

Single-plume luminescence postprocessing

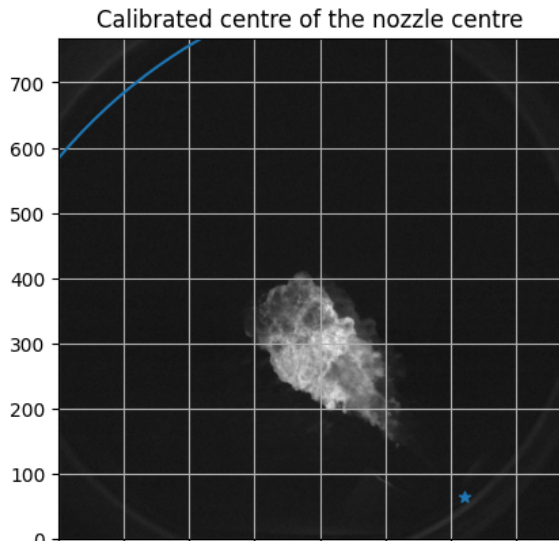
Data Analysis Pipeline

April 27, 2026

- **Luminescence imaging** captures visible light emitted by a burning fuel plume without an external laser.
- This module provides an automated pipeline for:
 - ① Preprocessing: Rotation, crop, bilateral filtering, and background subtraction.
 - ② Extraction: Creating Time-Distance (T-D) intensity maps.
 - ③ Analysis: Binarizing the T-D map to extract flame penetration and lift-off.
 - ④ Global Metrics: Computing frame-wise average intensity.
- GPU-accelerated via CuPy with automatic fallback to NumPy.

Calibration & Preprocessing

- Calibration centers the nozzle and masks out external regions using inner and outer radius parameters.
- The video is rotated so the plume extends horizontally.
- Bilateral filtering smooths the image while preserving edges.



Time-Distance Intensity Map

- Summing the foreground strip collapses each frame into a 1D intensity profile.
- Stacking these profiles creates a 2D image (Time vs. Distance).
- Brightness indicates integrated luminescence at that distance and time.

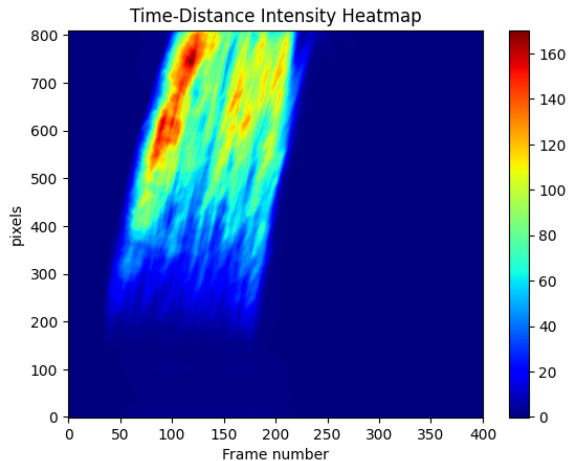
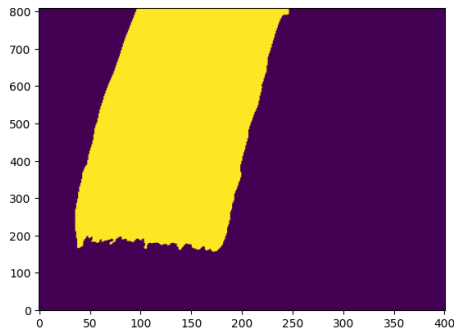


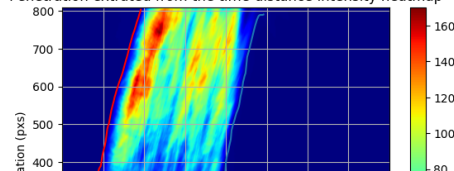
Figure: Time-Distance Intensity Heatmap

Binarization and Feature Extraction

- The T-D map is binarized using triangle thresholding.
- Isolated noise is removed by keeping the largest connected component.
- **Flame Penetration:** Farthest edge of the flame.
- **Lift-off:** Base of the visible flame.



Penetration extrated from the time-distance intensity heatmap



Frame-wise Average Intensity

- Summing all pixel values and dividing by pixel count gives a mean intensity per frame.
- Serves as a scalar proxy for overall combustion intensity.
- The filtered curve isolates the true plume signal from the background.

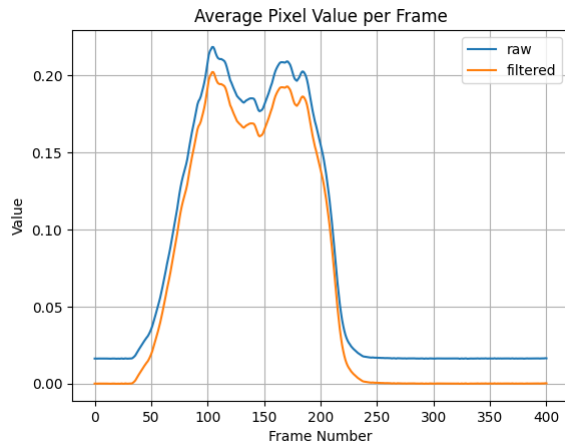


Figure: Average Pixel Value per Frame

Summary of Outputs

- **Video Strip (.avi)**: Rotated and filtered plume.
- **Numpy Archive (.npz)**: Data arrays for further processing.
- **Metrics (.csv)**: Per-frame average intensity, flame front, and lift-off.
- **Heatmap (.csv / .npz)**: Full Time-Distance map.
- All saving operations run asynchronously for performance.